

K-maps

A Karnaugh map lays truth-table ones on a grid so adjacent cells differ by one bit

Adjacent ones, bigger groups

- Two-variable maps use Gray order on both axes
- Three and four variables add more cells but the same rule: circle the largest legal groups
- A group of four drops two literals
- Don't-care X cells can join a group when they help
- Pick the minimal SOP when the challenge asks, one term per group

Browser lab

Digital Design and Verification Platform

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INTERACTIVE TOOL

K-map minimizer (2–6 vars)

Fill a Gray-coded Karnaugh map (0 / 1 / X). For 5–6 variables, MSB planes of 4-var maps sit side by side. Groups and a minimal sum-of-products update as you click.

Starter example: 2-variable XOR ($F = A'B + AB'$) with two groups highlighted.

Challenge 0 / 26 cleared

AND (2): Fill the map so $F = 1$ only when $A=1$ and $B=1$.

Show hint

Check

Next

Load challenge map

Idle

AND (2)

OR (2)

XOR (2)

XNOR (2)

NAND (2)

F = A

Pick SOP: XOR

Pick SOP: AND

AND (3)

Majority (3)

A ⊕ B (ignore C)

Pick SOP: majority

Decode 101

Σm(0,1,3)

Pick SOP: Σm(0,1,3)

Single minterm (4)

Quad group

Pick SOP: wrap quad

Don't cares fill

Pick SOP with X

Σm(1,5,9,13)

Pick SOP: C'D

5-var m0

5-var F=A

Pick SOP: F=A (5)

6-var m0

Karnaugh map

Load starter example

Clear map

VARIABLES

2 (A B) ▾

Minimal SOP

F = A'B + AB'

Σm(1,2) · 2 group(s)

A'R

AB'

K-map starter

Workbook practice

- In the workbook track, sketch a two-input XOR map by hand and write its SOP
- On three variables, group ones for at least two of A, B, or C
- Mark why top and bottom rows are neighbors on a four-var map
- Name one pitfall: grouping only in a straight line and missing a wrap-around quad

Pitfalls to watch

- Do not treat diagonal cells as adjacent, they are not
- A group must be a power-of-two size
- And remember: the browser lab is literacy
- Real logic minimization still needs consistent variable order and don't-care specs from

Your turn

- Complete the checklist for at least one track, preferably both
- In the browser, finish a few challenges after the starter
- On paper, draw one map and circle the groups
- When you are ready, take the short quiz, then continue to SOP and POS